

Figure 1.1: MULAG FME 500 front mower on a Unimog during roadside verge maintenance. Source: [MULAG Fahrzeugwerk 26].

experience, and randomising the simulator across a family of plants produces policies that transfer without being tuned to one machine. The second is that the actuator itself can be learned. Rather than deriving valve and cylinder behaviour from first principles and fitting its parameters, a network trained on recorded motion can represent the command-to-motion relationship directly, including the nonlinearities that analytical models approximate away.

Combining the two is attractive and not straightforward. A learned actuator model must be cheap enough to evaluate inside the training loop, which rules out the high-fidelity models used for offline analysis, and it must remain valid across the family of machines the policy will meet. This has motivated a line of work in which a compact code describing the current plant is inferred from recent interaction history and supplied to the controller so that a single policy can serve a family rather than a single machine. Such methods report substantial gains in legged locomotion and in manipulation, and they are the direct antecedent of the architecture examined here.

1.3 Challenges of Precise Tracking with Hydraulic Mowing Arms

A valve command does not produce a proportional cylinder motion.[1] The relationship is nonlinear, depends on load and supply pressure, and is subject to dead time between the command and any measurable response. It also differs between machines: seal condition, fluid temperature, hose compliance and the attachment fitted all change it, so two machines of the same model respond differently and one machine responds differently across a season. A controller that inverts only the geometry therefore leaves the largest source of tracking error unaddressed.